Table 1. Metadata for acoustical monitoring sites at Gates of the Arctic National Park and Preserve (GAAR). Dates reflect each site's recorded NVSPL deployment period(s).

Site	Site Name	Dates	Vegetation	Wilderness	Elevation (m)	Latitude	Longitude
GAARAMLA	NA		NA	NA	NA	NA	NA
GAARANAK	NA		NA	NA	NA	NA	NA
GAARARRI	NA		NA	NA	NA	NA	NA
GAARFLOR	NA		NA	NA	NA	NA	NA
GAARHAMM	NA		NA	NA	NA	NA	NA
GAARITKI	NA		NA	NA	NA	NA	NA
GAARKURU	NA		NA	NA	NA	NA	NA
GAARMWAL	NA		NA	NA	NA	NA	NA
GAARNARV	NA		NA	NA	NA	NA	NA
GAARNNUT	NA		NA	NA	NA	NA	NA
GAARNUTU	NA		NA	NA	NA	NA	NA
GAARNWAL	NA		NA	NA	NA	NA	NA
GAARPALA	NA		NA	NA	NA	NA	NA
GAARSELB	NA		NA	NA	NA	NA	NA
GAARUPNO	NA		NA	NA	NA	NA	NA
GAARWALK	NA		NA	NA	NA	NA	NA

Table 2. Summary of sound levels at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2022. For each site, includes the 90th percentile (LA10), existing (LA50), natural (LAnat), and 10th percentile (LA90) ambient sound levels (dB re 20 uPa, A−weighted broadband, 12.5−20,000 Hz) during the day (0700−1900) and night (1900−0700).

Time	SiteID	LA10	LA50	LAnat	LA90
Day	GAARAMLA	26.9	21.8	21.8	18.7
Night	GAARAMLA	27.7	24.3	24.3	22.1

Table 3. Number of days of audio analyzed for the Listening Center and SPLAT detection methods at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2022 − the basis for the analysis−period text reported in the Methods section.

Type	DaysText
ListeningCenter	NA
SPLAT	32

Table 4. Executive summary for Gates of the Arctic National Park and Preserve (GAAR), Summer 2022: mean overall noise audibility, top contributing noise sources, and ambient sound levels (LA50/LAnat, day and night) by site.

SiteID	Noise	Aircraft	NA 3	NA 4	LA50 Day	LAnat Day	LA50 Night	LAnat Night
GAARAMLA	0.1	0.1	NA	NA	21.8	21.8	24.3	24.3

Table 5. Noise impact (dB, existing minus natural ambient) and Listening Area Reduction (%) by site at Gates of the Arctic National Park and Preserve (GAAR), Summer 2022, for day, night, and the combined 24−hour period.

SiteID	DayImpact	DayLAR	NightImpact	NightLAR	AllImpact	AllLAR
GAARAMLA	0	0	0	0	0	0

Table 6. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2022.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARAMLA	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.7	0.7	0.0	0	0.0
GAARAMLA	1.1	1	Aircraft	1.1	Jet	0.2	0.1	0.2	143.8	27	100.0
GAARAMLA	1.2	1	Aircraft	1.2	Prop	0.3	0.0	0.3	298.4	172	285.8

Table 7. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2022.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	2423	1483	2310	3600	2718	3600	3042	3600	3600	3600	3600	3600	3600	GAARAMLA
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARAMLA
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARAMLA
Mean	3600	3600	3439	3392	3600	3600	3600	3600	3600	3600	3600	3270	2840	3186	3422	3199	3600	3120	3435	3439	3600	3306	3600	3600	GAARAMLA

Table 8. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2022.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARAMLA	No Sound Audible	0	0	0	0	0	0	0	0	0	0	0	0.0	0.0	0.0	0.0	0.0	0	0.0	0.0	0.0	0	0	0	0
GAARAMLA	Jet	0	0	0	0	0	0	0	0	0	0	0	0.1	0.2	0.0	0.2	0.1	0	0.1	0.0	0.0	0	0	0	0
GAARAMLA	Prop	0	0	0	0	0	0	0	0	0	0	0	0.1	0.8	0.9	0.0	0.1	0	0.2	0.2	0.1	0	0	0	0

Table 9. Percent time above sound level thresholds for acoustic records collected at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2022. Values show the percent of time ANS−weighted sound levels (20−1,250 Hz) and the full measured frequency range (12.5−20,000 Hz) exceeded 35, 45, 52, and 60 dB (LAeq,1s), by site and day/night.

Site ID	Frequency (Hz)	35 dB Day	45 dB Day	52 dB Day	60 dB Day	35 dB Night	45 dB Night	52 dB Night	60 dB Night
GAARAMLA	Full (12.5−20,000)	3.7	0.1	0	0	13.3	0.1	0	0
GAARAMLA	ANS (20−1,250)	2.0	0.0	0	0	9.2	0.0	0	0

Figure 1. GAARAMLA: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2022.

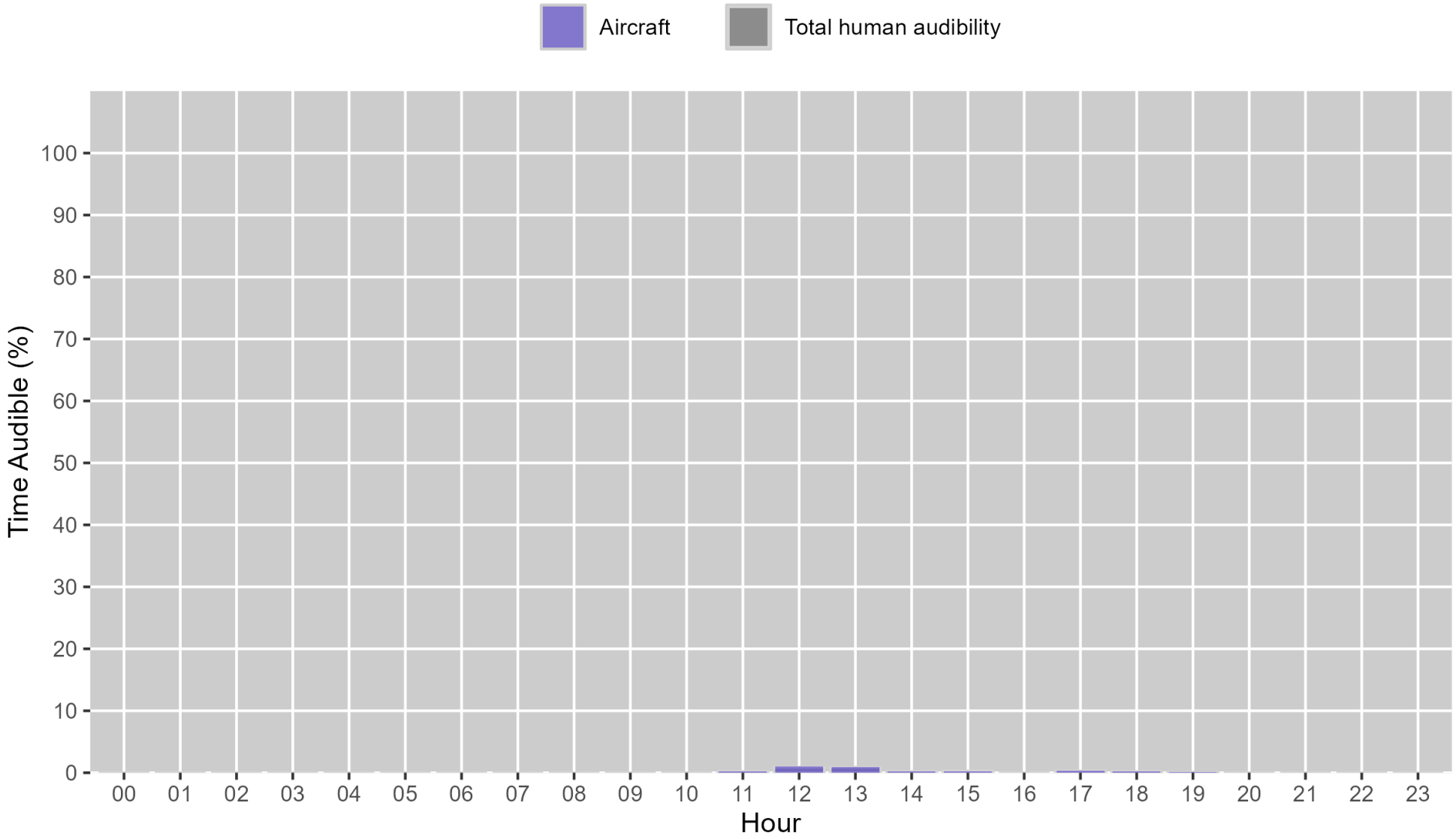


Figure 2. GAARAMLA: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2022.

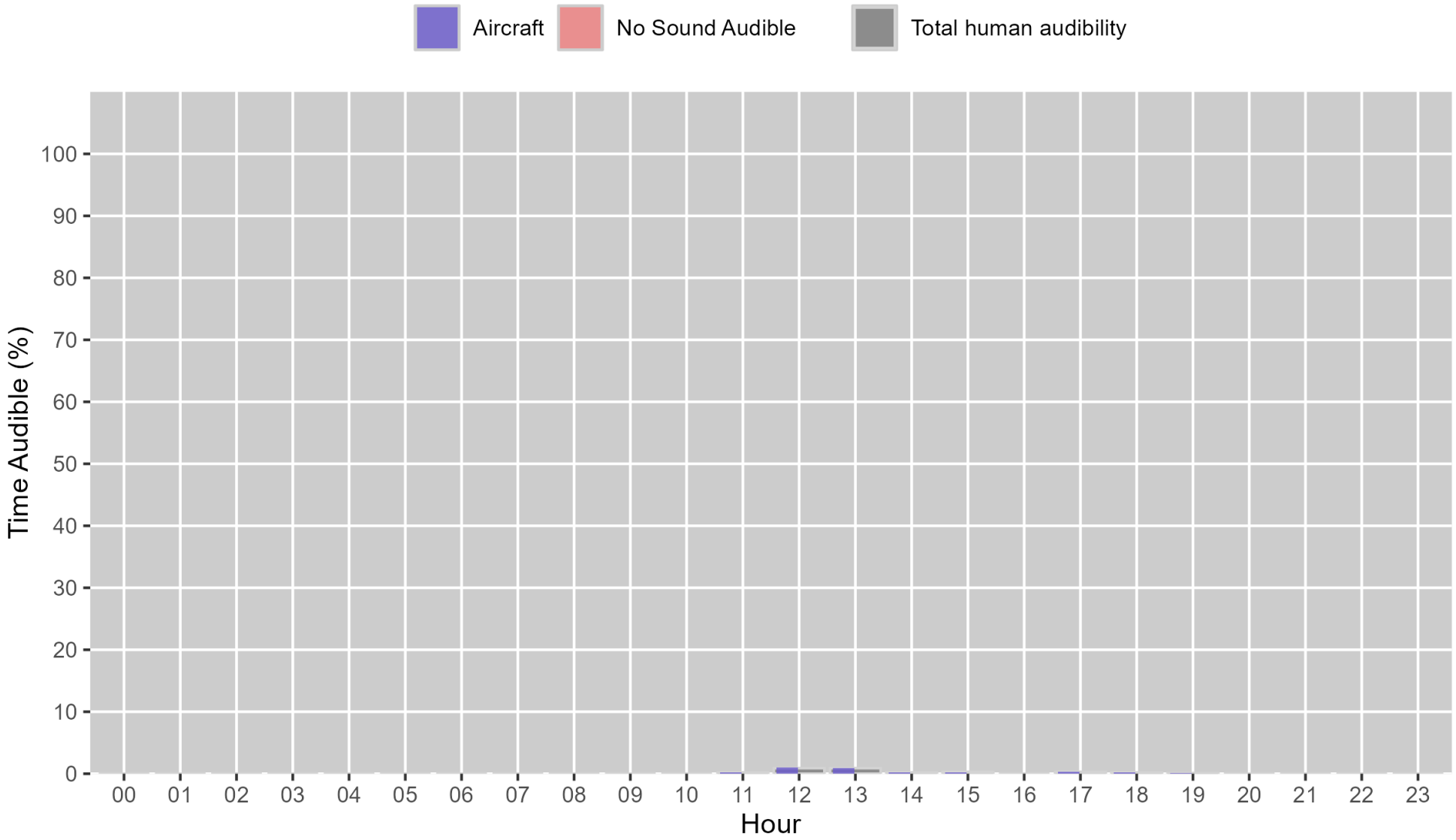


Figure 3. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARAMLA in Summer, 2022, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

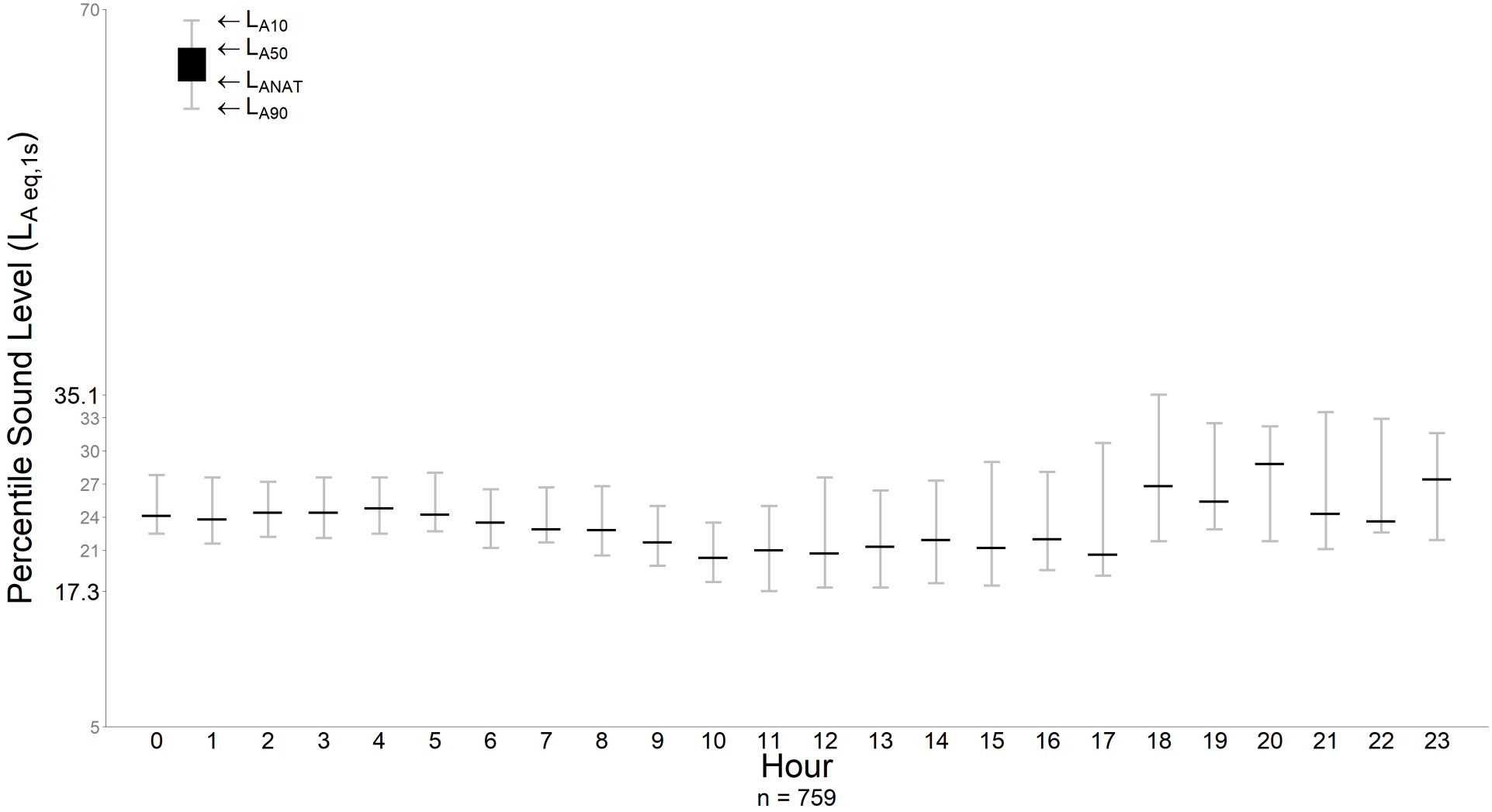


Figure 4. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARAMLA in Summer, 2022.

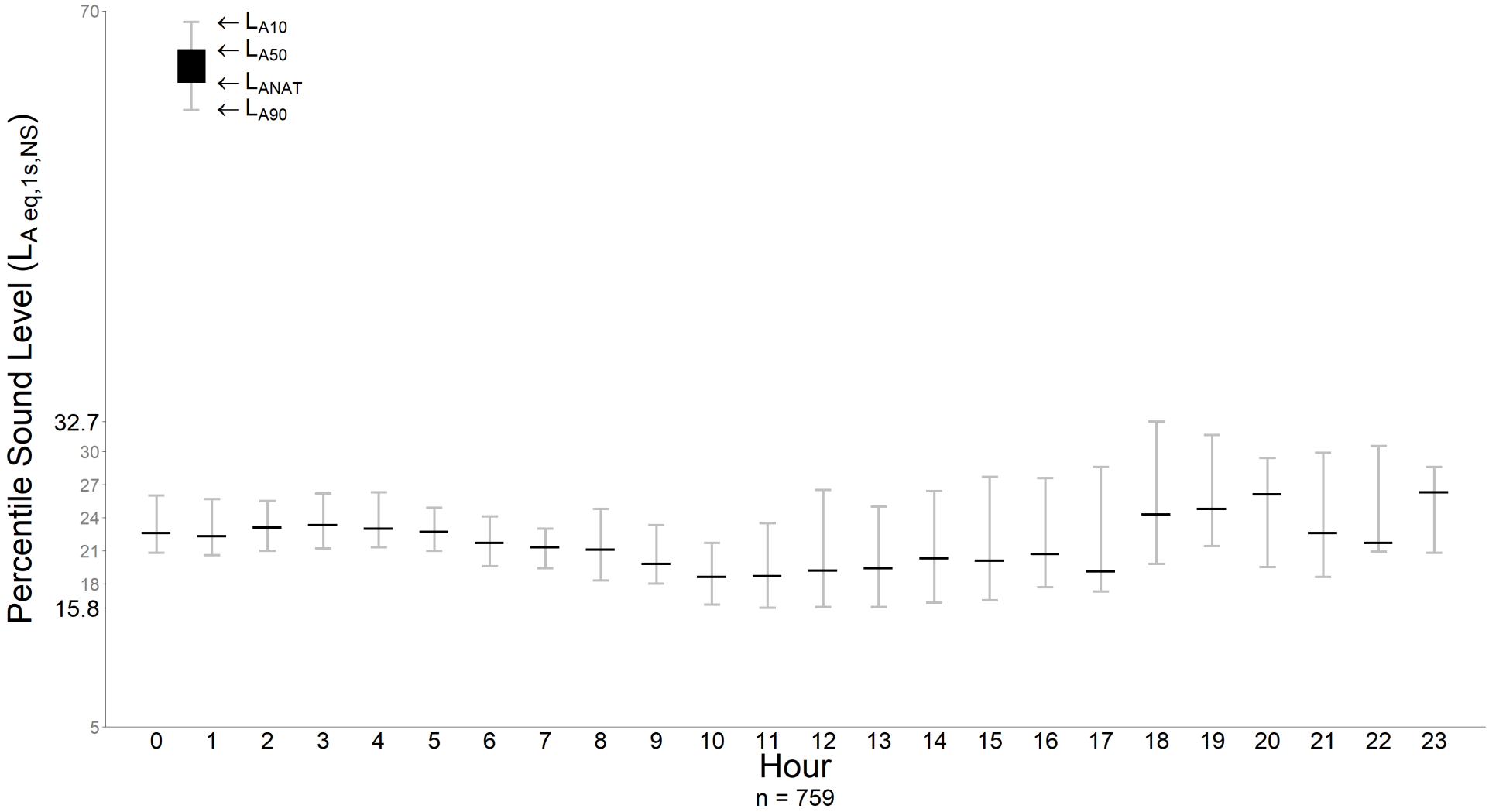


Figure 5. GAARAMLA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2022.

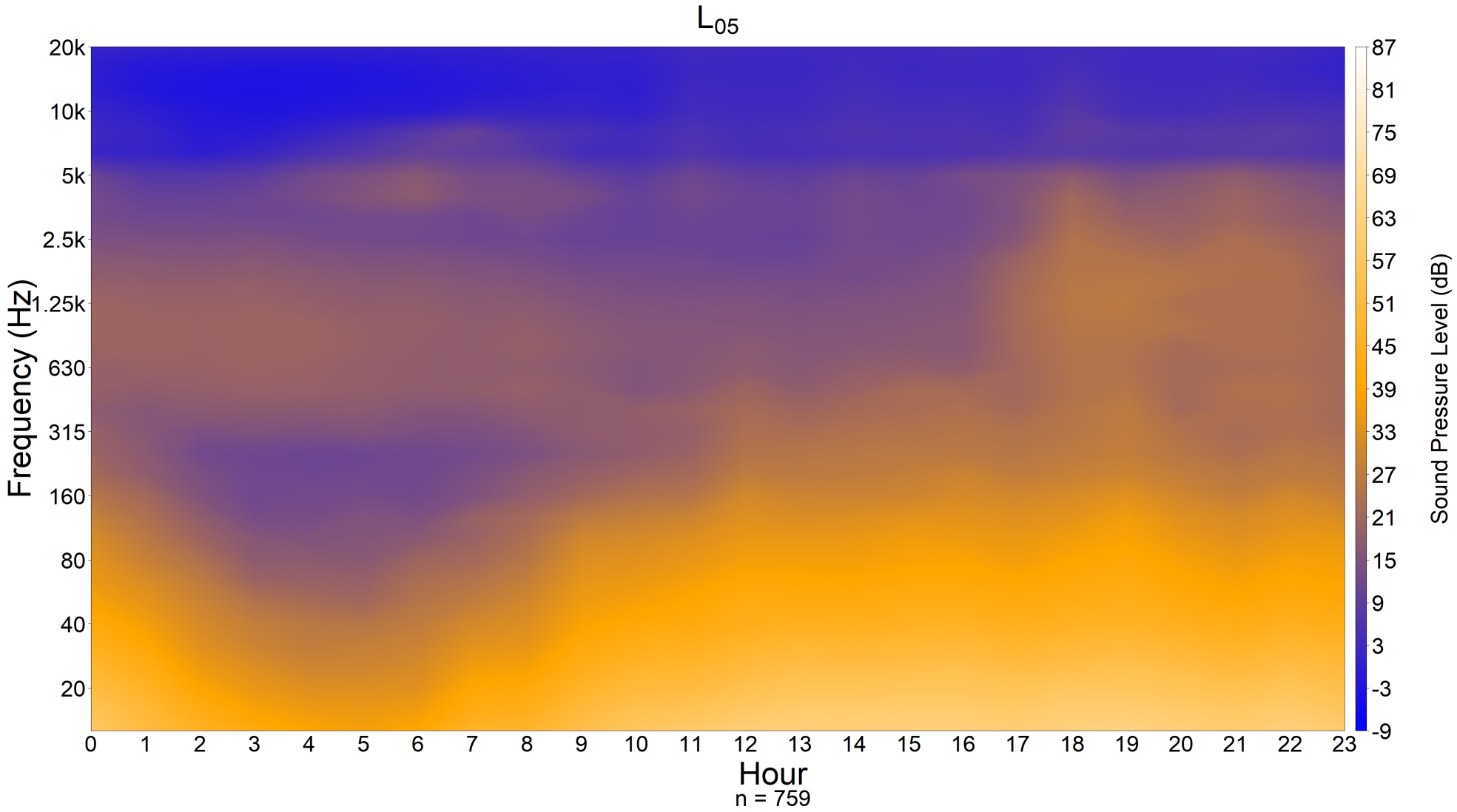


Figure 6. GAARAMLA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2022.

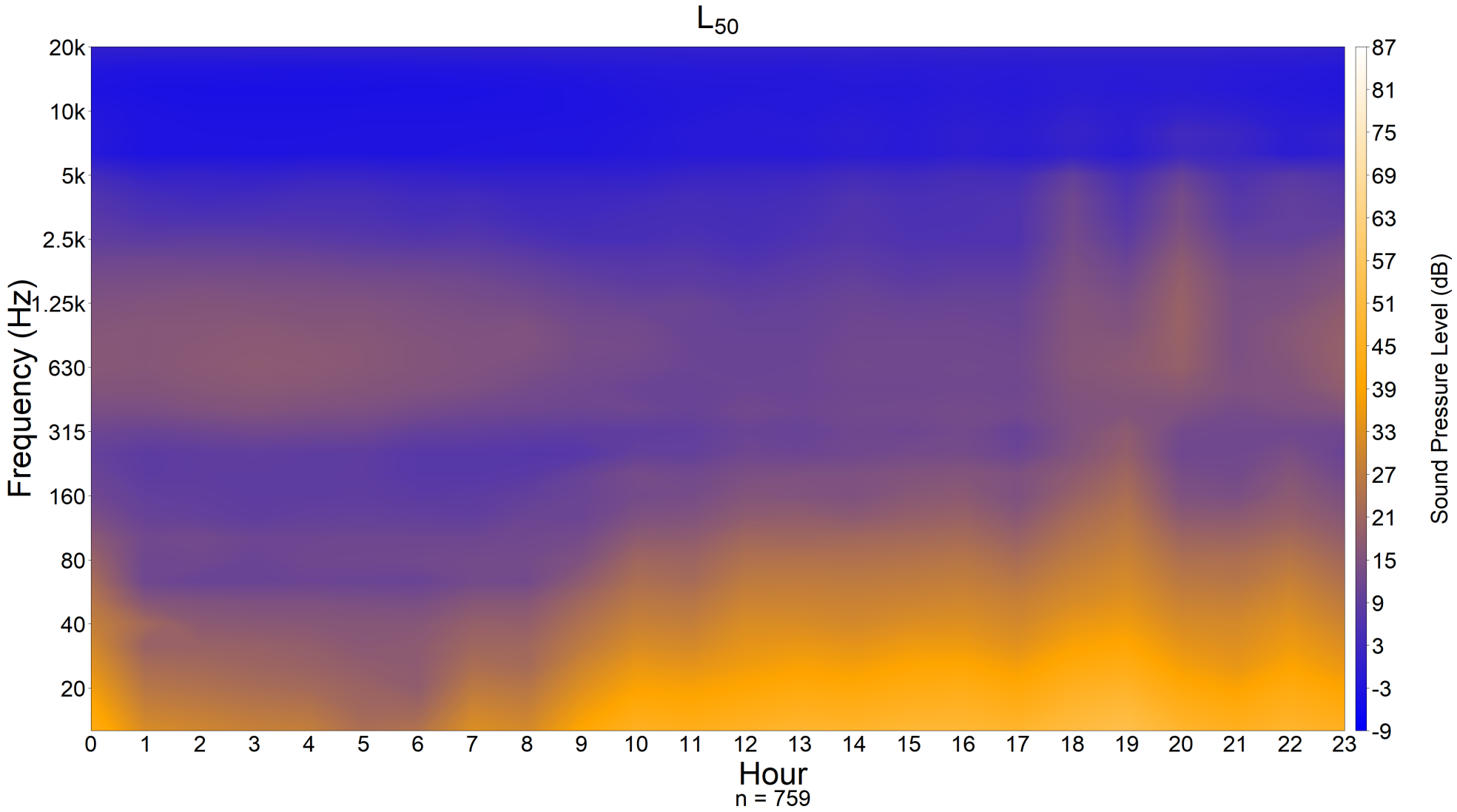


Figure 7. GAARAMLA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2022.

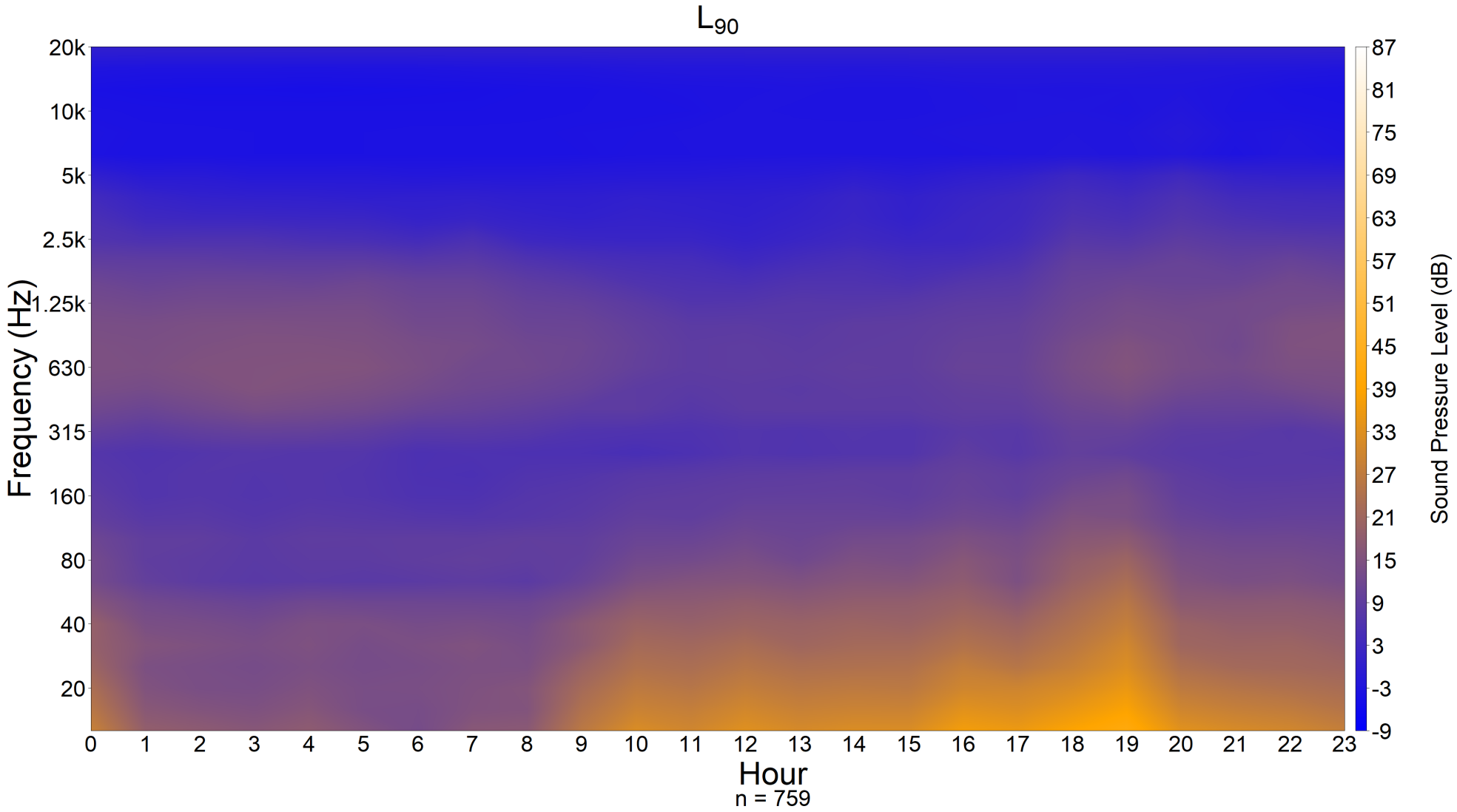


Figure 8. GAARAMLA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2022.

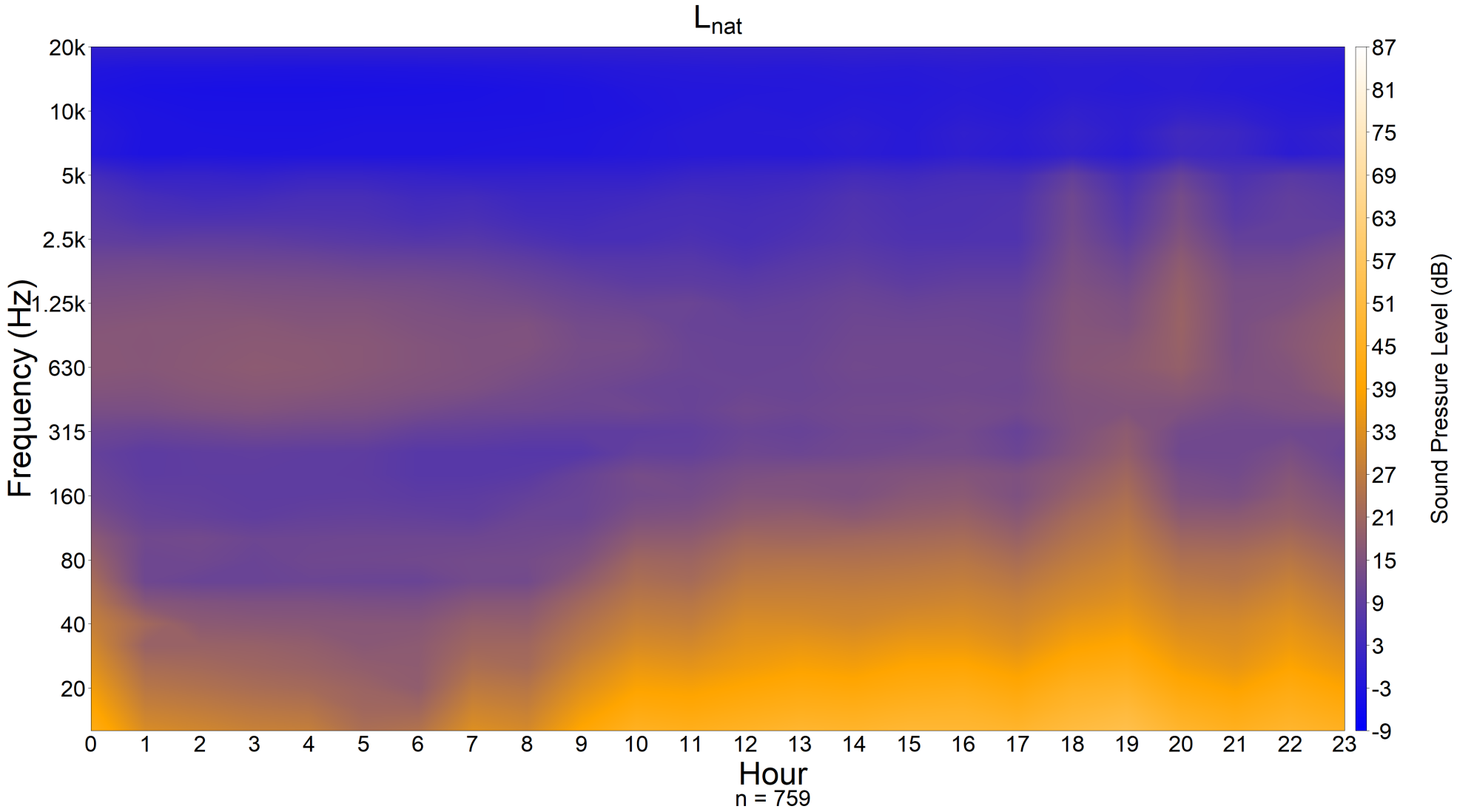


Figure 9. GAARAMLA: median and mean duration of noise−free intervals by hour of day, in Summer 2022.

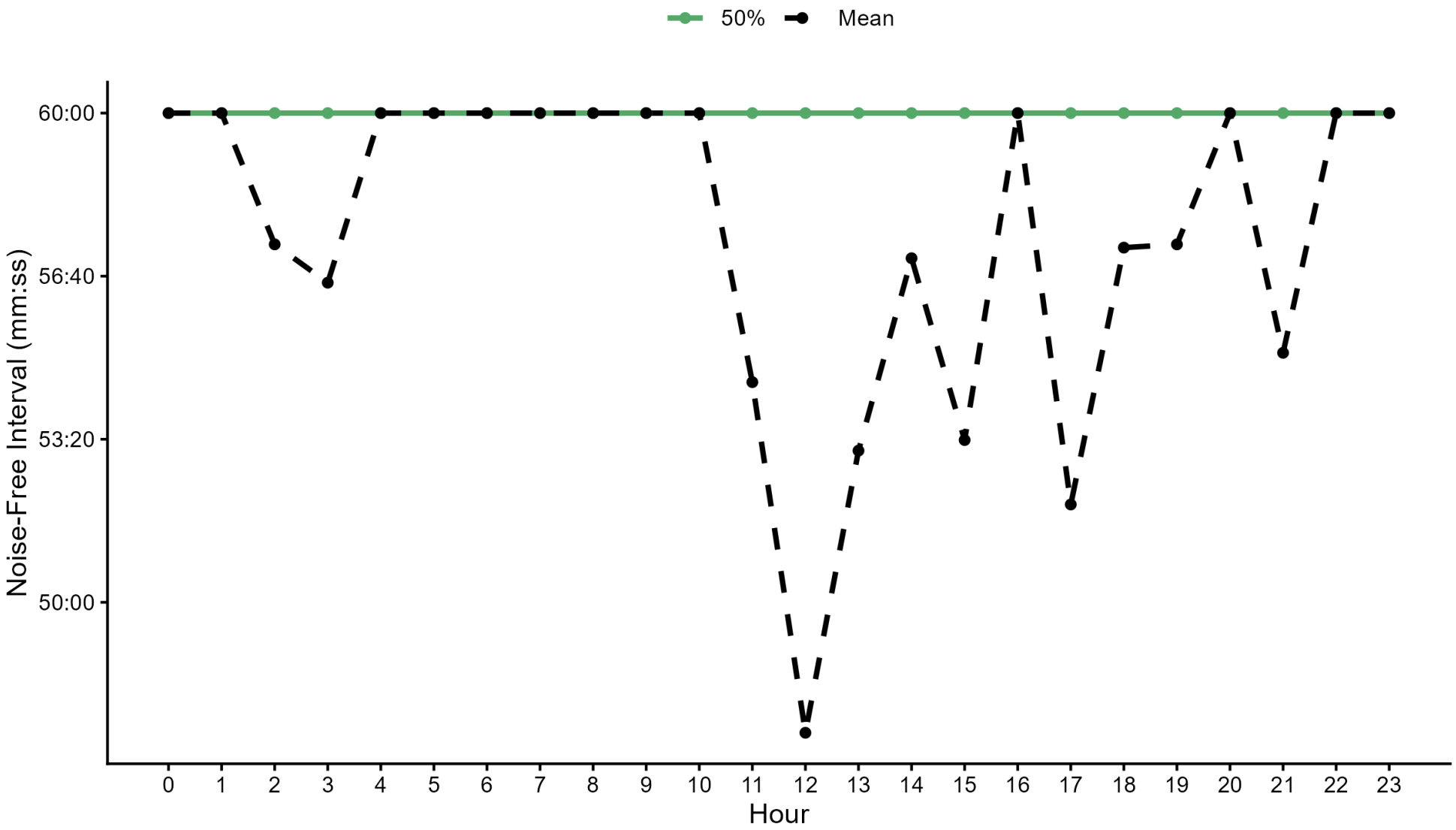


Figure 10. GAARAMLA: percent of time audible by sound source category (noise/natural), in Summer 2022.

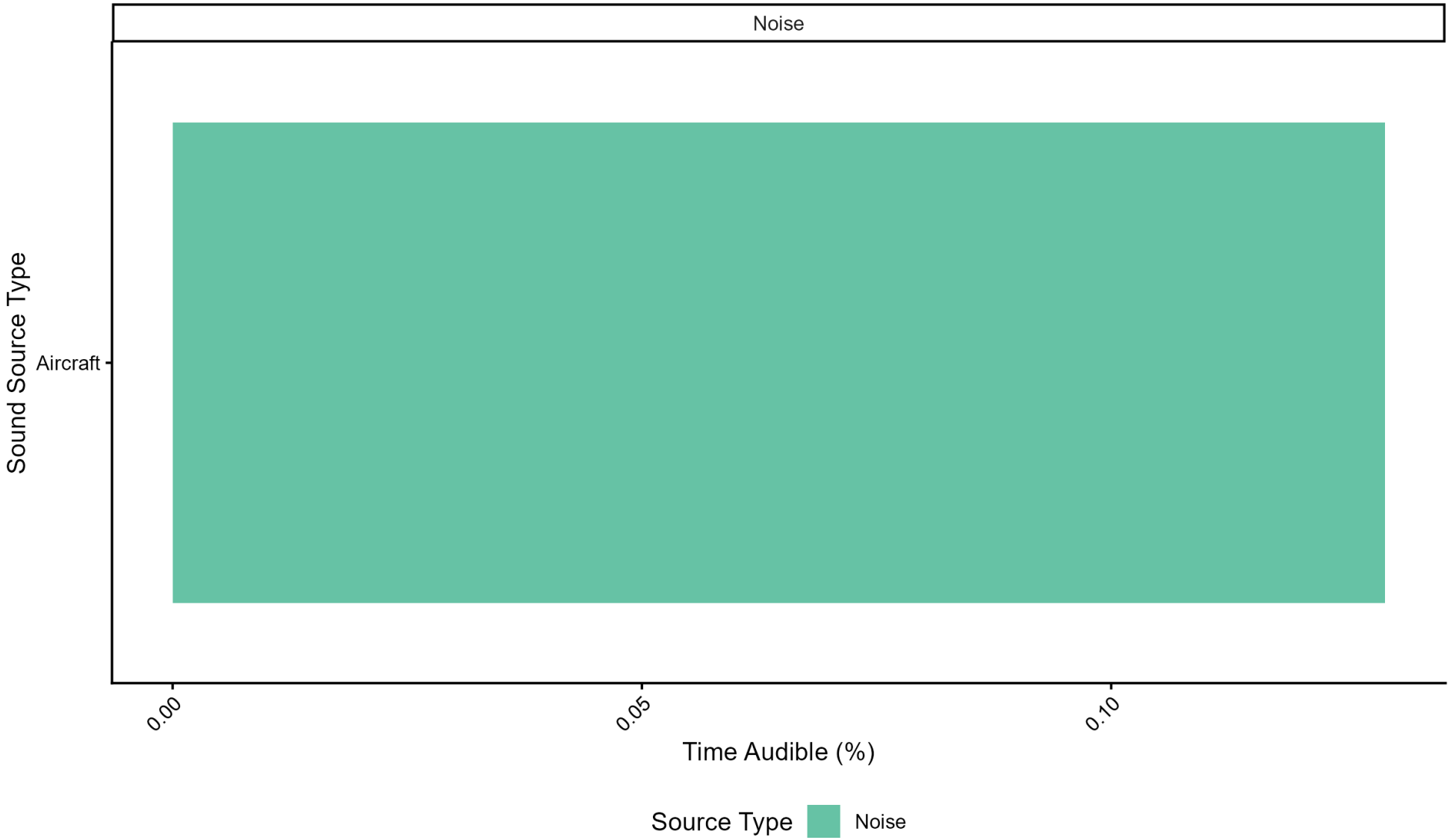


Figure 11. GAARAMLA: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2022.

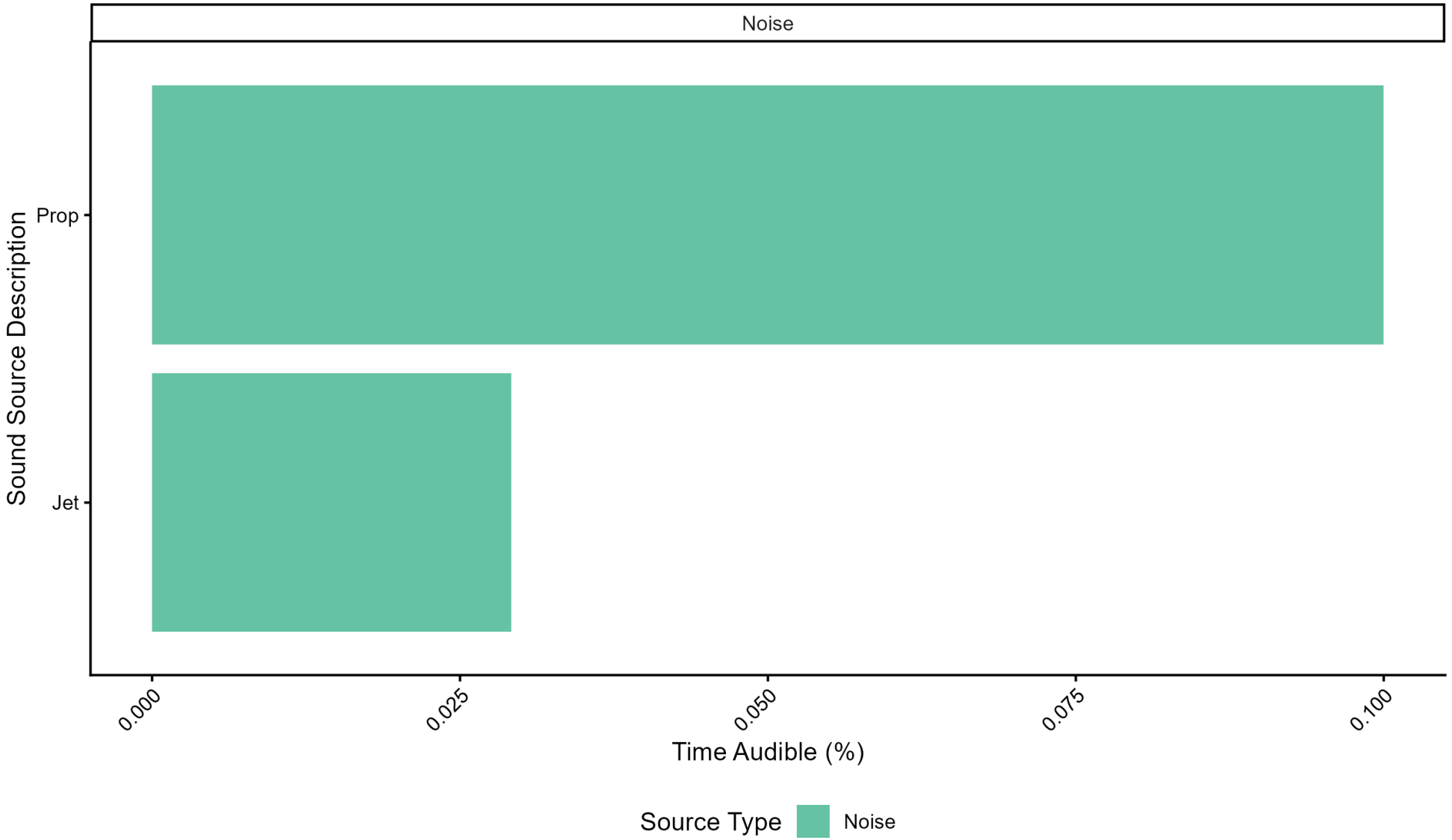


Figure 12. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARAMLA in Summer, 2022. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

